

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2024-2025

Comprehensive Examination
(EC-3 Makeup)

Course No. : DSECLZG522
Course Title : Big Data Systems
Nature of Exam : Open Book
Weightage : 40%
Duration : 2.5 Hours + 20 Mins
Date of Exam : 13-09-2025 (EN)

No. of Pages	= 3
No. of Questions	= 8

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q.1. A healthcare data analytics application utilizes a Cloud-based PostgreSQL cluster. The analytics team conducts patient record analysis on the data stored on the Cloud instance. It takes approximately 30 seconds per user query on average. You are tasked to implement a Memcached based cache in the healthcare center's data center to expedite these queries, aiming for an average query latency of 10 seconds. The Memcached cache has an access latency of 5 seconds. When the cache is established, data can only be transferred from the cache to the application and not directly from the Cloud DB.

What should be the projected hit rate in the cache to achieve this ambitious average query latency target?

What type of locality of reference is used here?

[Marks 5]

Q.2. Describe the most appropriate scheduling mechanism to be configured with Hadoop YARN in the following scenarios:

- (a). A company wants to run jobs of equal size on a Hadoop cluster. The number of jobs can vary at any point in time.
- (b). A company wants to run short jobs and long jobs on the same cluster and short jobs cannot always be starved by long jobs.
- (c) A company wants to divide the Hadoop cluster resources in different proportions between the departments and does not care about wastage of unutilized resources.
- (d) A company wants to run jobs of different nature concurrently on the same cluster and is worried about wastage of resources

[Marks 5]

Q.3. Suppose you are working with a large e-commerce database that contains customer orders, and you want to analyze the average order total for each customer in the year 2020. The 'orders' table has the following schema: order_id (int), customer_id (int), order_date (string in the format 'yyyy-mm-dd'), and order_total (double).

Using Hive, design a query that calculates the average order total for each customer, considering only the orders made in the year 2020. Be sure to include a description of how your query works and any assumptions you made about the data.

Additionally, discuss how you would optimize this query for performance if the 'orders' table contains millions of records, and explain the trade-offs between different optimization strategies.

[Marks 5]

Q.4. Suppose you're working on a project that involves analyzing a massive log file containing user interaction data from a web application, where the file size exceeds the available memory. Your task is to design an efficient data processing pipeline to extract specific user behavior patterns. Consider the trade-offs between using Resilient Distributed Datasets (RDDs), DataFrames (DF), and Datasets (DS) for this task. Describe a scenario where you would opt for RDDs over DataFrames or Datasets, and provide a detailed explanation of why RDDs are more suitable for this particular use case, including how you would leverage their characteristics to optimize data processing and minimize potential drawbacks.

[Marks 5]

Q.5. Consider the following code fragment and answer the questions below.

- (a) What are the transformation operations?
- (b) What are the action operations?
- (c) What will be the output of the first print statement?
- (d) What will be the output of the second print statement?

```
System.setProperty("hadoop.home.dir", "d:\big_data");
SparkConf conf = new
SparkConf().setAppName("MyFirstProgram").setMaster("local[*];
JavaSparkContext sc = new JavaSparkContext(conf);
JavaRDD<Integer> myRDD = sc.parallelize(Arrays.asList(1, 2,
3, 4, 5, 6, 7, 8, 9));
myRDD = myRDD.map(x -> x % 2 == 0 ? x * x : x * x * x);
System.out.println(myRDD.first()); // first print stmt

myRDD = myRDD.filter(x -> x > 100 && x < 500 ? true : false);
System.out.print(myRDD.count()); // second print stmt

List<Integer> list = myRDD.collect();
for (int i : list) {
    System.out.println(i);
}
sc.close();
```

[Marks 5]

- Q.6. In each of the scenarios given below, answer the corresponding question with brief but adequate reasoning.
- (a) A spark application is started on a cluster of 3 nodes from a client system using the spark-submit command. To kill the Spark application, the user performs a Ctrl+C on the client system. What is the expected behavior?
 - (b) You are designing a system leveraging Spark's ability to integrate with the Hive Metastore to analyze daily sales transactions belonging to an e-commerce company getting accumulated in an HDFS folder. It is not possible to copy or move the data files into any other storage location. Suggest a suitable type of Hive table to manage the data.
 - (c) You have to keep a set of RDDs in memory, but you need to look for CPU vs memory utilization traded-off because memory is limited. How could you achieve this as a Spark application developer?

[Marks 5]

- Q.7. Design a real-time data integration pipeline using Apache Spark to merge streams of data from multiple sources: social media, IoT sensors, and log files. The pipeline should handle different data formats (JSON, CSV, Avro), perform data cleansing, and write the integrated data to a unified data warehouse.

Describe the architecture of the data integration pipeline, including:

Data Ingestion: Reading data from various sources using Spark's built-in connectors (e.g., Kafka, Kinesis, File).

Data Processing: Handling different data formats, performing data cleansing (e.g., handling missing values, data normalization), and transforming data into a unified schema.

Data Integration: Merging data from multiple sources into a single, cohesive dataset.

Provide a high-level overview of how to implement this pipeline using Spark's APIs (in Scala or Python), including any necessary code snippets to illustrate key concepts.

[Marks 5]

- Q.8. You are working for a startup company that is building an e-commerce platform. They decided to use DynamoDB for storing the data. What are the advantages of using Provisioned mode, when compared with On-demand mode. In Provisioned mode, they anticipate the following workload requirements:

- Read Operations: 50,000 strongly consistent reads per second, each reading 8KB items.
- Write Operations: 20,000 transactional writes per second, each writing 3KB of data.

Calculate the number of RCU (Read Capacity Units) and WCU (Write Capacity Units) to be provisioned to meet the above requirements?

[Marks 5]
